

3A DAO Pyth Network Integration

Audit report prepared by 0xMilenov

1 About 0xMilenov

Independent Security Researcher

As a software engineer based in Bulgaria, 0xMilenov is a passionate expert in blockchain security, with a keen focus on **Web3, smart contracts, and vulnerability detection.**

Collaborating with renowned figures, he's contributed significantly to Web3 security projects and developed impactful tools.

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2 Protocol Summary

3A DAO Pyth Network Integration

This audit focused on the integration of **Pyth Network** price data for managing and converting token prices within the:

```
- OrchestratorPriceFeed.sol
- PricesOrchestratorPusher.sol
- PricesOrchestrator.sol
- ConvertedPriceFeed.sol
```

Usage flow:

```
1. Price retrieval and validation
2. Token price conversion
3. Emission of price update signals
4. Management of valid tokens and their prices
```

Checked possibile vulnerable places

```
- `Access control checks` (onlyOwner)
- `Price data validation and freshness`
- `Token address validation`
- `Conversion accuracy and consistency`
```

Specifics

```
- `OrchestratorPriceFeed`: Ensured the correct initialization and validation of oracle and token addresses, and verified the
↪  accuracy of price retrieval and updates.

- `PricesOrchestratorPusher`: Assessed batch price retrieval, conversion between USD and EUR, and validation of token
↪  addresses. Verified the update and management of valid tokens and their prices.

- `PricesOrchestrator`: Evaluated the integration with Pyth Network, update of price IDs, and management of valid time
↪  ranges for price data.

- `ConvertedPriceFeed`: Verified the correct conversion of primary and secondary price feeds.
```

During the audit of the Pyth Network Integration contracts, no vulnerabilities were identified.

Disclaimer: This security review does not guarantee against a hack. It is a snapshot in time of {X} according to the specific commit. Any modifications to the code will require a new security review.

	Impact: High	Impact: Medium	Impact: Low
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

3 Risk Classification

3.1 Impact

- High - leads to a loss of a significant portion (>10%) of assets in the protocol, or significant harm to a majority of users.
- Medium - global losses <10% or losses to only a subset of users, but still unacceptable.
- Low - losses will be annoying but bearable--applies to things like griefing attacks that can be easily repaired or even gas inefficiencies.

3.2 Likelihood

- High - almost certain to happen, easy to perform, or not easy but highly incentivized
- Medium - only conditionally possible or incentivized, but still relatively likely
- Low - requires stars to align, or little-to-no incentive

3.3 Action required for severity levels

- Critical - Must fix as soon as possible (if already deployed)
- High - Must fix (before deployment if not already deployed)
- Medium - Should fix
- Low - Could fix

4 Executive Summary

Overview

Project	3A DAO Pyth Network Integration
Repository	PythNetwork-Integration
Commit	...
Date	June 2024

Issues Found

Severity	Count
Critical Risk	0
High Risk	0

Medium Risk	0
Low Risk	0
Informational	0
Gas Optimizations	0
Total Issues	**0**